

MOHSIN MANAF

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Summary

Mechanical Engineering graduate with hands-on internship experience in technical documentation, maintenance operations, and engineering design. Proficient in preparing and interpreting technical manuals and engineering drawings, with expertise in CAD software such as SolidWorks and Fusion 360. Demonstrated ability to lead projects involving product design, failure analysis, and process optimization using data-driven approaches. Strong communication and coordination skills developed through cross-functional teamwork in shipbuilding and locomotive maintenance environments. Keen interest in aircraft maintenance engineering and configuration management, with a commitment to quality standards and regulatory compliance.

Education

T.K.M. COLLEGE OF ENGINEERING

Bachelor of Technology in Mechanical Engineering

Nov. 2021 – May. 2025

Kollam, Kerala

T.K.M. Centenary Public School

12Th Standard

June. 2019 – March. 2020

Kollam, Kerala

Trinity Lyceum, Nanthirickal

10Th Standard

June. 2017 – March. 2018

Kollam, Kerala

Relevant Coursework

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|--------------------|--------------------------|--------------------------|-----------------------|
| • Product Design | • CAD/CAM | • Operation Management | • Solid Works |
| • Vehicle Dynamics | • CFD using Ansys Fluent | • Additive Manufacturing | • Product Development |

Internships

Cochin Shipyard Ltd

Summer Intern

August 2024

Kochi, Kerala

- Acquired hands-on experience in the shipbuilding and assembly processes..
- Gained knowledge of ship systems, including safety ,propulsion and electrical systems.
- Assisted in the integration of various ship components, such as hull sections and machinery.
- Observed and supported quality control procedures to ensure structural and operational integrity of vessels.

Diesel Loco Shed, Ernakulam

Summer Intern

May 2023

Ernakulam, Kerala

- Assisted in monthly maintenance of diesel locomotives..
- Performed inspections and diagnostics on diesel engines and systems.
- Supported maintenance of traction motors and electrical systems.
- Gained exposure to locomotive assembly, including wheels, chassis, and suspension.

Projects

Design and Fabrication of a Tri wheeler in Association with Govt.of Kerala. | *Project Head*

March 2023

- Led the fabrication of an electrically powered assistive tricycle for wheelchair users.
- Designed a system for wheelchairs to lock onto and climb the high-clearance tricycle.
- Collaborated with teams to ensure safety and functionality of the vehicle.
- Managed fabrication, from materials procurement to assembly, meeting project deadlines.

XLR8 FORMULA STUDENTS CAR | *Steering Head, Driver*

July 2024

- Led the design and development of the steering system for a student Formula car and optimized vehicle dynamics
- Participated in Formula Imperial, Formula Bharat, and Supra competitions and won appreciable All India Rankings.
- Collaborated with cross-functional teams for seamless system integration.
- Conducted simulations and physical testing to validate steering performance under dynamic conditions.
- Managed timelines, resource allocation, and documentation for design reviews and technical presentations.

Design Modification and Failure Analysis of Screw Implant | *Fusion360, ANSYS*

January 2025

- Developed and modified 3D models of screw implants using Fusion 360 to enhance performance.
- Conducted Finite Element Analysis (FEA) in ANSYS to evaluate stress distribution and fatigue life.
- Identified potential failure points and iteratively refined designs based on simulation results.
- Benchmarked improved designs against standard implant geometries for comparative analysis.

Impact of printing parameters on the tribological behaviour of 3D printed components.

March 2025

- Investigated the tribological behavior of 3D printed components (PLA, ABS, Nylon) by conducting Pin-on-Disk wear tests, tensile, and hardness testing.
- Applied Design of Experiments (DOE) to study the impact of printing parameters (print speed, layer height, infill density) on wear and friction performance.
- Analyzed experimental data to optimize process parameters for improved mechanical properties and wear resistance.
- Documented findings in a detailed technical report, demonstrating strong skills in data analysis, materials characterization, and technical writing.

Industrial visit

Hindustan Machine Tools Limited

September 2022

- * Gained exposure to the manufacturing and assembly processes of CNC machines, lathes, and milling tools.
- * Observed quality control and precision inspection methods used in tool production.
- * Interacted with engineers to understand machine tool design and automation integration.
- * Learned about lean manufacturing practices and preventive maintenance techniques for high-precision equipment.

Technical Skills

Programming Language: Python, C

Design and Analysis Software: Solidworks, Fusion 360, Ansys Fluent

Other Software: Excel, PowerPoint, Minitab

Certification

NPTL Online Certification - Product Design and Development

July - August 2022

IIT, Roorkee

- Completed an in-depth course covering the full product development cycle from concept generation to market launch.
- Applied engineering and market-based approaches to develop user-centric, innovative product solutions.
- Gained knowledge in need identification, idea screening, prototyping, and design for manufacturability (DFM).
- Analyzed real-world case studies to understand practical applications of design strategies across various industries.

Leadership / Extracurricular

- Vehicle Dynamics and Steering Department Head of XLR8 Students Formula Team 2024
- Programme Committee Head of TEDx TKMCE and Logistics Head of MUNSOC TKMCE.
- Publicity Coordinator of ETIS'2024.
- Alumni Wing Coordinator for STEPS TKMCE
- Technical head of Film and Drama Club.
- NSS Volunteer and Blood Cell Coordinator.